

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Microprocessor Programming & Interfacing	Course Code:	EE303
	Program:	BS(EE)	Semester:	Fall 2018
	Duration:	180 Minutes	Total Marks:	100
	Paper Date:	21-DEC-18	Weightage	50%
	Section:	ALL	Page(s):	2
	Exam Type:	Final Exam		

Student : Name:

Roll No.

Section:

Instruction/Notes: It is an OPEN BOOK/OPEN NOTES exam, Show complete working for each question, direct answers are not acceptable. Make assumptions if you think some data is missing, but clearly mention your assumption. Attempt each question only once.

Question#1:

[20 Marks] CLO-02

Define a string of variable length in ROM of 89C52. Read the string and store index value, in default Stack, at which the character "a" has been found. Moreover, count the total occurrences of character "a" found in the stored string.

Your code must be generic. Use of loops is necessary; otherwise, ZERO credit will be given.

Question#2:

[20 Marks] CLO- 03

Write an assembly language program that generates Fibonacci series for 'n' a 4-bit number given as input via port P1, and displays each element in series one by one on 2x16 LCD every second after clearing the previous value. Formula to generate Fibonacci series is given below:

$$F[n] = F[n - 1] + F[n - 2]$$

Where,

$$F[0] = F[1] = 1$$

The user will prepare input number over binary switches and then gives a positive pulse (0-1-0) on control switch connected to P1.4. Once the user give input i.e., value of 'n', the program will display each element in series starting from F_0 to F_n . Example:

If $n = 15$ then,

1,1,2,3,5,8,...,987 series will be displayed

Your code must be as generic as possible. The switches introduce bouncing for 25msec. Assume that subroutines LCD_Command_Write, LCD_Data_Write, Delay_25msec are already given to you. Accumulator (A) is used to send command to LCD in LCD_Command_Write subroutine. Accumulator (A) is used to write data to LCD in LCD_Data_Write subroutine. Use the available subroutines in your solution.

Question#3:

[20 Marks] CLO- 04

There is a limited car parking space in a shopping mall. Maximum 8500 cars can be parked. An automatic vehicle security barrier is installed and it generates a short pulse whenever the vehicle is passed through the barrier. In order to determine whether the parking lot is full or not an application is required to be designed to switch on the buzzer (connected at pin P2.3) for 5 seconds at the entrance of 8500th car. Assume crystal frequency to be 6MHz. **NOTE:** Use on chip module of 89C52 microcontroller and write an interrupt driven code for the required system.

Question#4:

[40 Marks] CLO - 05, PLO5 (SI, O2)

You are required to design an automatic swimming pool water level maintenance system. This system is installed at swimming pool of the house. Water level can be detected by using two sensors (SN1 and SN2) connected at two points (minimum level and maximum level) of the pool. If the water level crosses the maximum level, sensor SN1 is activated and gives a pulse as output. If the water level crosses the minimum level, sensor SN2 is activated and gives a pulse as output.

The system comprises of two controllers. One controller detects the overflow or empty situation of the pool and sends a message to another microcontroller over serial interface (with baud rate 9600 bps) accordingly. On detection of overflow, the first microcontroller should send "O" to second microcontroller. On detection of water level below the minimum threshold, the first microcontroller should send "L" to second microcontroller.

On receiving the message "L", second microcontroller should start the DC motor.

On receiving the message "O", second microcontroller should stop the DC motor and ring the buzzer for 1 second with 125Hz frequency.

You are provided with following components

- 8051 microcontroller MC1 driver circuit with $F_{osc} = 12\text{MHz}$
- 8051 microcontroller MC2 driver circuit with $F_{osc} = 22\text{MHz}$
- 8051 microcontroller MC3 driver circuit with $F_{osc} = 12\text{MHz}$
- Water level sensor gives high to low pulse on detecting the level below the threshold
- Water level sensor gives low to high pulse on detecting the level below the threshold
- Water level sensor gives high to low pulse on detecting the level above the threshold
- Water level sensor gives low to high pulse on detecting the level above the threshold
- Buzzer

Things to do:

1. Chose the components according to the given scenario and make a block diagram of the system. Mention the component name you are using e.g., MC1, MC2, etc. [5 Marks]
2. Considering the block diagram you have made, identify the inputs and outputs of the system. [5 Marks]
3. Decide and mention which *modules* of 8051 you will use. Do this for both microcontrollers. [5 marks]
4. Write complete code for *both* microcontrollers [10+10=20 marks]
5. Dry run your code for two different inputs and proof correctness of your code. [5 Marks]